OPERATING SYSTEMS

UNIT V – FILE SYSTEM INTERFACE, IMPLEMENTATION & PROTECTION

SECTION A – 2-MARK QUESTIONS (3-MARK ANSWER FORMAT)

Answers, examples and simple diagrams

Q1. List the attributes of a file.

Definition:

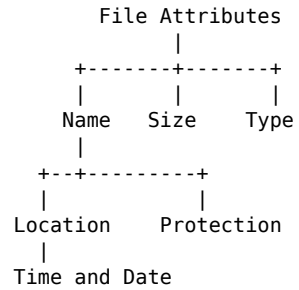
File Attributes are the properties or details of a file maintained by the Operating System.

Explanation:

- Name: Name of the file.
- Identifier: Unique number used to identify the file.
- Type: Type of file, such as PDF, JPG or TXT.
- Size: Amount of space occupied by the file.
- Location: Where the file is stored.
- Protection: Specifies who can read, write or execute the file.
- Time and Date: File creation and modification time.

Example: A file named notes.pdf has a name, size, type and location.

Diagram:



Q2. Define Free Space Management.

Definition:

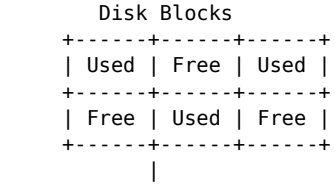
Free Space Management is a method used by the Operating System to keep track of unused space in a disk.

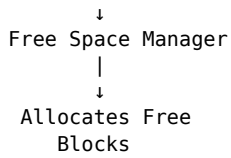
Explanation:

- It identifies empty disk blocks.
- It keeps a record of available space.
- When a new file is created, free blocks are allocated to it.

Example: Like keeping track of empty seats in a classroom.

Diagram:





Q3. What is a Directory Structure in File Systems?

Definition:

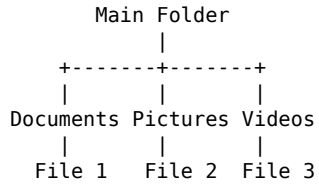
A Directory Structure is a method used by the Operating System to organize files and folders on a computer.

Explanation:

- It stores file names and related information.
- It organizes files into folders.
- It helps users find files easily.

Example: In a computer, we create folders like Documents, Pictures and Videos.

Diagram:



Q4. State the goal of Protection in an Operating System.

Definition:

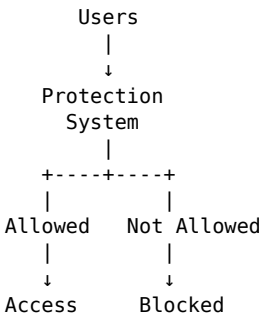
Protection is a mechanism used by the Operating System to control access to computer resources.

Explanation:

- Prevent Unauthorized Access: Stops unknown users from accessing files.
- Data Security: Protects important information.
- Controlled Access: Allows users to access only permitted resources.

Example: A password protects our personal files from other users.

Diagram:



Q5. What is an Access Matrix in Protection?

Definition:

An Access Matrix is a table used by the Operating System to define which user or process can access which resource.

Explanation:

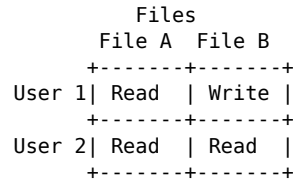
- Rows represent users or processes.
- Columns represent files or resources.
- Each cell contains permissions like Read, Write and Execute.

Remember:

R = Read W = Write X = Execute

Example: A student can read a file but cannot modify it.

Diagram:



Q6. Define File Mounting.

Definition:

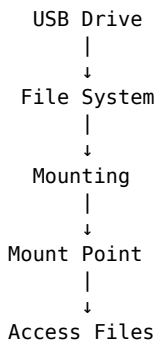
File Mounting is the process of making a file system available for use by attaching it to a directory called a mount point.

Explanation:

- The OS connects a storage device or file system.
- It attaches the file system to a directory.
- Users can access its files through that directory.

Example: When we connect a USB drive to a computer, its files become accessible through the system.

Diagram:



Q7. List Different File Access Methods.

Definition:

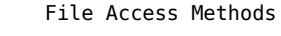
File Access Methods are techniques used to read or write data from a file.

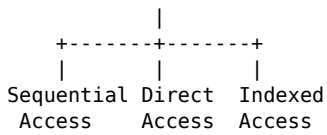
Explanation:

- Sequential Access: Data is accessed one by one in order.
- Direct Access: Data can be accessed directly from any required position.
- Indexed Access: An index is used to find the required data quickly.

Example: Sequential: Reading a book page by page. Direct: Opening a specific page directly. Indexed: Finding a topic using a book index.

Diagram:





Unit V – Quick Revision

Q.No	Topic	Important Keywords
Q1	File Attributes	Name, Size, Type, Location
Q2	Free Space Management	Unused Disk Blocks
Q3	Directory Structure	Files, Folders
Q4	Protection	Security, Permissions
Q5	Access Matrix	Users, Files, Permissions
Q6	File Mounting	File System, Mount Point
Q7	File Access Methods	Sequential, Direct, Indexed